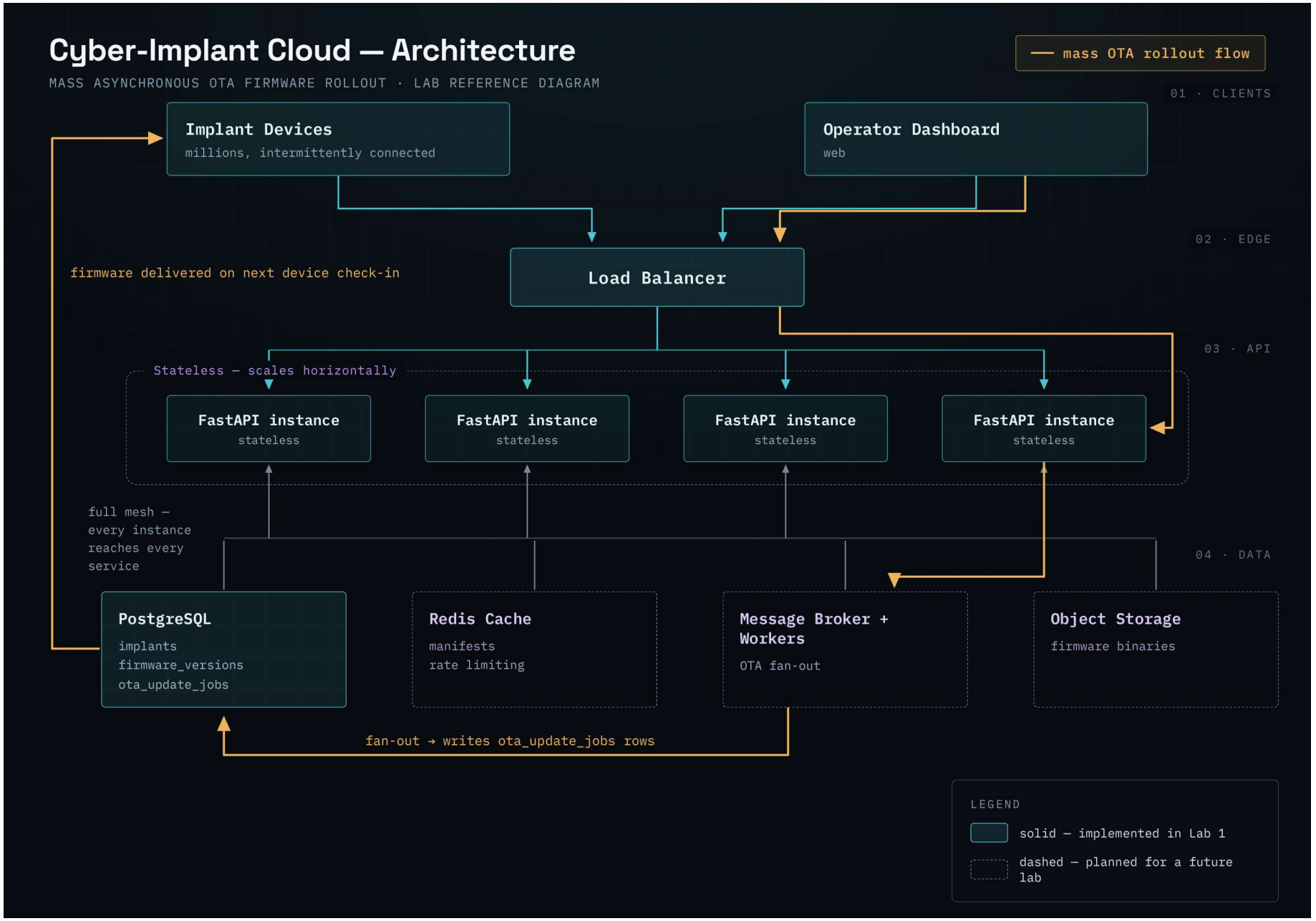

Cyber-Implant Cloud — Architecture
MASS ASYNCHRONOUS OTA FIRMWARE ROLLOUT · LAB REFERENCE DIAGRAM
— mass OTA rollout flow
01 · CLIENTS
Implant Devices
millions, intermittently connected
Operator Dashboard
web
02 · EDGE
firmware delivered on next device check-in
Load Balancer
03 · API
Stateless — scales horizontally
FastAPI instance
stateless
FastAPI instance
stateless
FastAPI instance
stateless
FastAPI instance
stateless
full mesh — every instance reaches every service
04 · DATA
PostgreSQL
implants
firmware_versions
ota_update_jobs
Redis Cache
manifests
rate limiting
Message Broker + Workers
OTA fan-out
Object Storage
firmware binaries
fan-out → writes ota_update_jobs rows
LEGEND
solid — implemented in Lab 1
dashed — planned for a future lab